

LC 921: Minimum Add to Make Parentheses Valid

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Problem Understanding

A parentheses string is valid if and only if:

- It is empty string
- **AB**, where A and B are both valid
- **(A)**, where A is a valid string

Example:

`() → ( ) → (())`

** not valid, need to make valid*

Here we need to o/p min additions to make it valid.

Intuition

(→ need a future closing), so **increase open**

) → if open > 0 , match with existing (, **decrease open**

else) has no matching (

→ add a (

→ **ans++**

C++ Implementation

```
class Solution {
public:
    int minAddToMakeValid(string s) {

        int ans = 0;
        int open = 0;

        for(char c : s) {
            if(c == '(') {
                open++;
            }
            else {
                if(open > 0) {
                    open--;
                }
                else {
                    ans++;
                }
            }
        }

        ans += open;
        return ans;
    }
};
```